

Making cuMES faster

Spend less time on each GPU pass.

Then need fewer passes to reach equilibrium.

CUDA execution

convergence trajectory

commit-backed measurements

RTX 4090 · ITERATION LATENCY

—19.1%

W7-X · 660.81 → 534.53 μ s/pass

V1.1 → V1.2 · CONVERGENCE

—25.4%

W7-X · 5,505 → 4,106 passes

Two levers, two kinds of evidence

$$T_{\text{run}} = T_{\text{setup}} + \sum_g (N_g \bar{t}_g + T_{\text{transfer},g}) + T_{\text{output}}$$

01 · EXECUTION

Reduce submission gaps, redundant transfers, synchronization, and kernel latency. Hold the numerical problem and controller decisions fixed.

02 · TRAJECTORY

Change cold seeds, time-step policy, vacuum handover, and radial transfer. Validate the new equilibrium and count every stage.

The audit starts at overhaul closeout

Milestone	Commit / date	What this deck covers
Design closure	dc0d0c4 · Aug 17	Starting boundary of the 197-commit inventory
Final reader closeout	56aa1a4 · Aug 18	Safety follow-ups complete; frozen trajectories retained
v1.0	dbb0f8e · Aug 26	CUDA graphs, Jacobian reduction, inverse mapping
v1.1	f7036ab · Aug 30	Finest-grid capture; Makegrid; zero-copy host bridge
v1.2	17867d5 · Aug 31	Recovery, seeds, handover, cubic/B-spline transfer
Later CUDA work	750d6a4 → 194415d	Concurrency, coarse QA/QH seeds, float accuracy/cache

Read every number with its measurement scope

Evidence	Protocol	Interpretation
Archived RTX 4090	CUDA 12.9; precise double; 50 warmup + 300 timed; six alternating graph pairs	Per-pass wall medians; frozen final-state hashes
Archived convergence	ADR-0007...0012; case-specific paired runs	Effective passes and final residuals; clocks sometimes noisy
New TITAN Xp	CUDA 12.1; sm_61; same source-pinned builds; six fixed triples	Before / optimized-direct / optimized-graph
New release comparison	Five timed alternating pairs after one warmup pair	CLI wall time includes startup and native output
Structural accounting	Counts and byte sizes derived from implementation	Explicitly not a measured wall-time speedup

Faster execution

01 Keep the equilibrium method. Improve the way work reaches the GPU and data reaches the host.

The one-fence loop was the foundation

ALREADY COMPLETED DURING OVERHAUL

Resource	Before	After
Ordinary host barriers / pass	5	1
cuFFT workspace · W7-X	≈ 10.4 MB	≈ 5.5 MB
Seed slab copy / zeroing	6 + 6 calls	1 + 1 calls

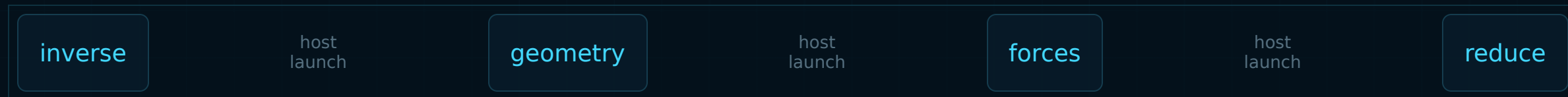
THE CONTROL FENCE CARRIES THE DECISION

GPU reductions write one control record. The host checks geometry, residuals, convergence, and restart policy after the same fence.

Timing events and telemetry use that existing boundary; the iteration does not stop just to print.

A graph removes launch gaps, not the control decision

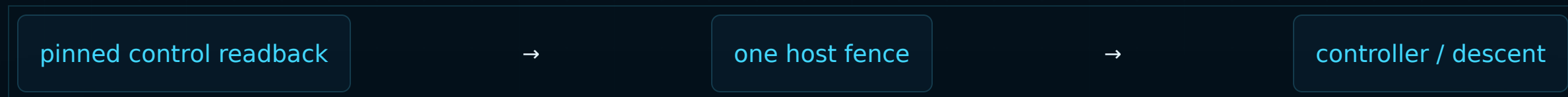
DIRECT STREAM



GRAPH REPLAY



BOTH PATHS



Fixed-boundary schedules can be replayed. Free-boundary vacuum coupling and verification dumps retain direct execution.

The retained CUDA bundle: RTX 4090

SOLOVEV · $n_s = 55$



28.8% **less wall time per pass** · 1.404 $\times$ throughput

W7 - X · $n_s = 99$



19.1% **less wall time per pass** · 1.236 $\times$ throughput

Parallelize the Jacobian validity reduction

W7-X KERNEL MEASUREMENT



14.54× **kernel speedup** · 93.1% less kernel time

KEEP THE VALIDITY SEMANTICS

One serially loaded block becomes up to 128 blocks of 256 threads, followed by one final reduction.

Preserved result	Implementation
Minimum oriented $\sqrt{g}$	Tie-break by lowest linear index
Maximum $ \sqrt{g} $	Include sign-flipped samples
Non-finite count	Accumulate across all blocks

Expose all θ points to the GPU

BEFORE

$$(\theta/2, \zeta_{\text{tile}}) \times 2 \text{ serial } \theta \text{ passes}$$

One lane synthesizes a point in each poloidal half.

AFTER

$$(\theta, \zeta_{\text{tile}}) \times 1 \text{ point per lane}$$

The modal accumulation order for each output is retained. The block policy caps the total thread count.

W7-X has 30 θ points: the θ dimension grows from 15 to 30 lanes. Its isolated latency gain was not separately recorded.

Production graph caching earns its complexity

RTX 4090 · precise double	Direct stream	Graph replay	Wall reduction
Solovev · $n_s = 55$	117.14 μ s	82.33 μ s	29.7%
W7-X · $n_s = 99$	562.10 μ s	534.53 μ s	4.9%

NUMERICAL GATE

Final-state hashes match graph on/off:
3c09fd3260b003a8 · Solovev
ce46a7cbe693601a · W7-X

MEASUREMENT

1,000-pass preheat; six alternating pairs; each run discards 50 warmup passes and times 300. No graph-node argument mutation.

Reproduced now: the same CUDA commit on Pascal

TITAN Xp	7c6508a · before	5379fca · direct	5379fca · graph	Bundle reduction
Solovev · $n_s = 55$	162.66 μs	165.16 μs	152.38 μs	6.32%
W7-X · $n_s = 99$	1739.59 μs	1573.82 μs	1561.67 μs	10.23%

Every before/direct/graph repetition has the same final-state hash within each workload.

Keep the timing spread visible

New fixed-iteration runs	Median $\pm$ MAD $\cdot \mu\text{s}$	Median run $p_{95} \cdot \mu\text{s}$	Range of run medians $\cdot \mu\text{s}$
solovev $\cdot$ before	162.66 $\pm$ 0.89	177.19	154.07–163.67
solovev $\cdot$ direct	165.16 $\pm$ 0.89	175.69	155.89–166.29
solovev $\cdot$ graph	152.38 $\pm$ 0.59	241.82	136.19–153.20
w7x $\cdot$ before	1739.59 $\pm$ 0.55	1752.91	1736.18–1740.17
w7x $\cdot$ direct	1573.82 $\pm$ 0.60	1591.89	1572.76–1576.04
w7x $\cdot$ graph	1561.67 $\pm$ 0.56	1854.88	1560.99–1564.79

MAD and range describe six run medians. The p_{95} column is the median of six within-run p_{95} values; it is a different statistic.

Stop copying results that cannot be published

FINEST-GRID-ONLY FIELD CAPTURE

Three-grid solve	Before	After
Derived-field evaluations	3	1
Coarse-stage capture calls	2	0

The two intermediate snapshots would be overwritten. Their field evaluation and device-to-host capture are skipped.

BOOZER HOST VIEW

6 **spectral families** + 7 **half-grid fields** are exposed as spans into the existing host snapshot.

This avoids another owning host copy and the native-file round trip. It does not remove the solver's device-to-host publication.

Makegrid: improve one-time host work

FLAT RESPONSE STORAGE · V1.0

Three contiguous component arrays; coil-major blocks, R -fast indexing. Removes nested vector traversal and simplifies direct in-memory response handoff.

933b80a → vacuum-field be6d6cd

PARALLEL GENERATION · V1.1

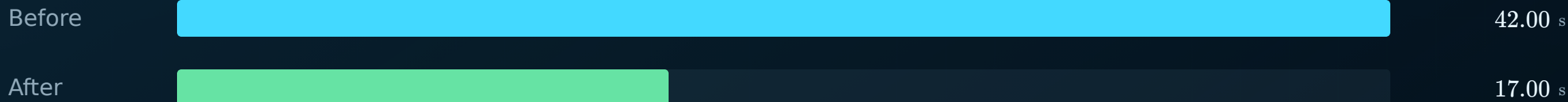
Independent grid points run on CPU workers. Dynamic chunks contain 8 **items**; useful worker count is capped at roughly 1 **per** 32 **items**.

VFIELD_MAKEGRID_THREADS=1 selects serial execution.

No isolated timing is recorded for either commit. These changes affect construction, not the per-iteration vacuum update.

Supporting tools also became faster

FIGURE GENERATION



Recorded wall time: 59.5% **lower** · 2.47× speedup.

CHANGES IN 5310769

Batched 2-D IFFT replaces a direct mode sum, checked to 10^{-14} . Figure rendering uses separate processes and copy-on-write input data.

The commit also reports $\approx 100\times$ synthesis acceleration, without a reproducible timing protocol.

B-spline setup belongs off the critical path

W7-X transfer	Direct host batch	Host matrix construction	Uploaded map
33 → 66	1.90 ms	36.7 μ s	17,424 bytes
66 → 99	2.94 ms	136.5 μ s	52,272 bytes

ONE REUSABLE MAP, 936 PROFILES

Apply the map to all six spectral families on the GPU. The spectral state stays device-resident.

OVERLAP WITH STAGE 1

Both maps take 173.3 μ s in total; one average W7-X device iteration was $\approx$ 462 μ s. The coarse stage takes 1,315 **iterations**.

Later: coordinate capture, overlap independent solves

Eight finite-difference columns	Serial	Two workers	Throughput
QA	2.823 s	1.247 s	2.26×
QH	2.250 s	1.025 s	2.20×

One process-wide host mutex protects graph capture against a concurrent device-wide stage fence. Ordinary graph execution and the iteration loop remain unlocked.

Measured execution ideas that did not ship

Experiment	Measurement	Decision
Split R/Z and λ forces · overhaul context	Registers 108 $\rightarrow$ 82/54; 1.20–1.45 $\times$ slower	Reject: duplicated input traffic
Forward tile 4 · later CUDA	Graph-on W7-X 1.573 $\rightarrow$ 1.518 ms; 3.5%	Below 5% gate
Inverse pack remapping	Five alternating runs; flat to slower	Restore original mapping
Derive poloidal tables from sin/cos	$\approx$ 0.5% slower; hash unchanged	Remove variant
Tile 4 · four-worker QA / QH	0.814 $\rightarrow$ 0.801 s / 0.678 $\rightarrow$ 0.682 s	Only 1.6% / – 0.7%

Resource reuse had a small upper bound

Eight-solve aggregate	QA	QH
Multigrid wall	3.1525 s	2.6627 s
Stage setup	0.13690 s · 4.3%	0.14062 s · 5.3%
Iteration	2.89850 s · 91.9%	2.40979 s · 90.5%
Derived-field capture	0.06035 s	0.06185 s
Stage teardown	0.04174 s	0.03798 s
Other orchestration	0.01505 s	0.01246 s

Even deleting all setup and teardown gives only a 5.6%/6.7% ceiling, while retaining ≈ 69 MB per W7-X worker. Caching was rejected without implementation.

Fewer iterations

02 Change the starting state and continuation path. Keep the force tolerances and validity gates explicit.

W7-X: account for every saved stage

Policy after each change	$n_s = 33$	$n_s = 66$	$n_s = 99$	Total	Saved vs prior
v1.1 reference	1,877	1,617	2,011	5,505	—
Step recovery	1,741	1,568	1,635	4,944	561
Shaped cold start	1,315	1,559	1,633	4,507	437
Catmull-Rom transfer	1,315	1,443	1,402	4,160	347
B-spline · v1.2	1,315	1,419	1,372	4,106	54

1,399 fewer full passes: 25.41% less work. This is a cumulative sequence; the individual gains are conditional on earlier changes.

Recover a step reduced by the early transient

ONE CONSERVATIVE ATTEMPT

$$\Delta t \leftarrow \min(\Delta t_0, 1.1\Delta t)$$

Wait for 250 **accepted passes since the latest restart**. Attempt recovery once per fixed-boundary stage.

W7-X · SINGLE $n_s = 99$

Controller	Passes	FSQR
Reference	2,953	9.924×10^{-13}
Recovery	2,711	9.988×10^{-13}

242 **fewer passes** · 8.20%

Recovery scales across multigrid stages

W7-X grid	Reference	Recovery	Fewer passes
33	1,877	1,741	136
66	1,617	1,568	49
99	2,011	1,635	376
Total	5,505	4,944	561 · 10.19%

ARCHIVED RTX WALL MEDIANS

Single grid: 2.0583 → 1.9217 s (6.63%).
Multigrid: 3.3109 → 3.0021 s (9.33%).

VALIDATION

Final multigrid residuals:
 $R\ 9.989 \times 10^{-13} \cdot Z\ 1.590 \times 10^{-13} \cdot \lambda\ 5.116 \times 10^{-14}$.
Final-grid checkpoint replay: 1 **pass**.

Shape the interior; preserve axis order and LCFS

$$w_m(s) = s^{m/2} [1 + c(1 - s)]$$

Fixed 3-D cold start	Envelope c	Before → after	Reduction
W7-X multigrid	0.12	4,944 → 4,507	8.84%
W7-X single · first policy	0.12	2,711 → 2,627	3.10%
W7-X single · tuned policy	0.129	2,711 → 2,465	9.07%

At $s = 1$ the correction is exactly one. Near the axis it retains the $s^{m/2}$ order. The host seed changes once; the per-pass GPU workload is unchanged.

Axisymmetric tuning has three separate contributions

Solovev policy	Stage trajectory	Total	Reduction vs 906
Reference	251 → 199 → 456	906	—
Coarse envelope −0.07	241 → 199 → 455	895	1.21%
Plus initial-step scaling	238 → 193 → 387	818	9.71%
Plus geometric λ predictor	235 → 193 → 387	815	10.04%

For configured $\Delta t = 0.9$: single-grid start 1.05; first continuation grid 1.02; prolonged grid 1.08.

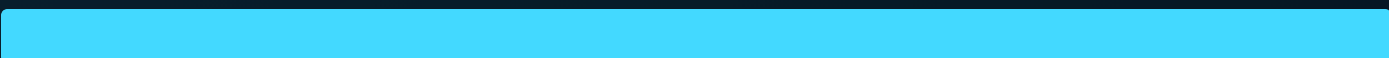
A geometric λ seed helps fine cold starts

STRAIGHT-FIELD-LINE PREDICTOR

$$q = \sqrt{g}/R^2, \quad \lambda_\theta = 1 - q/\langle q \rangle_\theta$$

Project the derivative on the host. Fixed-boundary starts use 65% of the predictor. Invalid geometry falls back to zero λ .

SOLOVEV · COLD $n_s = 55$

Reference		533.00 passes
Step only		416.00 passes
Step + λ		354.00 passes

33.58% **fewer passes** before any multigrid transfer.

Free-boundary predictors need separate qualification

Precise-double case	Reference	Seeded	Seed / predictor	Seeded FSQR
CTH-like · $n_s = 15$	489	384	$c = 0.12$	9.932×10^{-11}
CTH-like · $15 \rightarrow 25$	592	563	$c = 0.12$	9.942×10^{-11}
W7-X · $n_s = 51$	1,831	1,797	$c = 0.03$	9.705×10^{-13}
Solovev · $16 \rightarrow 32$	1,047	1,025	λ scale=1.0	9.822×10^{-15}

Using $c = 0.12$ on fine W7-X regressed to 1,953 passes. The smaller fine-grid correction is a measured stability choice.

Activate the physical vacuum force earlier

FSQR + FSQZ < 3 × 10⁻² instead of 10⁻³

Free-boundary case	Reference	Seeded	+ coarse step	+ earlier vacuum
CTH-like · single	489	384	347	309
CTH-like · multigrid	592	563	446	431
W7-X · n _s = 51	1,831	1,797	1,797	1,733
Solovev · multigrid	1,047	1,025	1,025	1,025

The coarse 3-D step multiplier is 17/14: CTH-like starts at 0.85 instead of 0.7. Activation retains the existing reset and validation sequence.

Free-boundary wall measurements follow the pass counts

TITAN Xp · archived	Reference median	Tuned median	Reduction	Pairs
CTH-like · 15 → 25	1.93 s	1.50 s	22.3%	7
W7-X · $n_s = 51$	8.37 s	7.93 s	5.3%	5

CTH-LIKE SPREAD

Reference: 1.91–2.01 s
Tuned: 1.48–1.50 s
Identical 592/431 counts in every pair.

W7-X SPREAD

Reference: 8.32–8.39 s
Tuned: 7.85–7.97 s
Identical 1,831/1,733 counts in every pair.

A better transfer saves work on refined grids

Linear → Catmull-Rom	Before	After	Fewer passes	Final FSQR
W7-X · fixed	4,507	4,160	347 · 7.70%	9.986×10^{-13}
Solovev · fixed	815	766	49 · 6.01%	9.695×10^{-17}
CTH-like · free	431	424	7 · 1.62%	9.920×10^{-11}

Four coarse samples retain curvature that two-point interpolation discards. The first stage is unchanged; only the refined-stage starting state changes.

Global B-splines improve the transfer again

Catmull-Rom → B-spline	Before stage counts	After stage counts	Total saved	Final FSQR
W7-X · fixed	1315 → 1443 → 1402	1315 → 1419 → 1372	54 · 1.30%	9.997×10^{-13}
Solovev · fixed	235 → 190 → 341	235 → 193 → 326	12 · 1.57%	9.973×10^{-17}

NUMERICAL IMPROVEMENT

An interpolating cubic B-spline uses the converged radial profile globally. The GPU map matches the direct host B-spline trajectory.

EXECUTION IMPROVEMENT

One precomputed map applies to every spectral profile. Asynchronous preparation hides its host construction behind the coarse solve.

Use the transfer that each solve class qualified

Solve class	Selected transfer	Evidence for limiting scope
Precise-double fixed boundary	Global cubic B-spline	W7-X 4106; Solovev 754
Precise-double 3-D free boundary	Catmull-Rom	CTH: spline 424 → 425, so reject
Axisymmetric free boundary	Linear	Cubic 1025 → 1045; spline →1074
Mixed-float	Linear	Higher-order transfer not selected
Fixed without spline dependency	Catmull-Rom fallback	Explicit build/runtime opt-outs retained

Reproduced now: exact release convergence counts

TITAN Xp · precise double	v1.1 passes	v1.2 passes	Reduction	v1.2 FSQR
Solovev · 5/11/55	906	754	16.78%	9.972×10^{-17}
W7-X · 33/66/99	5,505	4,106	25.41%	9.997×10^{-13}
Solovev · $n_s = 55$	533	354	33.58%	9.835×10^{-17}
W7-X · $n_s = 99$	2,953	2,465	16.53%	9.959×10^{-13}

All five timed repeats agree on every stage count and final printed residual triple. The input JSON is identical between versions.

Reproduced now: complete CLI wall time

New TITAN Xp · five pairs	v1.1 median ± MAD · s	v1.2 median ± MAD · s	Reduction
Solovev · multigrid	0.4651 ± 0.0018	0.4432 ± 0.0039	4.71%
W7-X · multigrid	7.2590 ± 0.0191	5.5994 ± 0.0183	22.86%
Solovev · single	0.4044 ± 0.0152	0.4385 ± 0.0138	−8.43%
W7-X · single	6.1740 ± 0.6661	5.3135 ± 0.0956	13.94%

Includes process startup, stage construction, iteration, final scientific fields, and native file output. Warmup runs and checkpoint gates are excluded from these five timed pairs.

Solovev wall estimates are inconclusive: confidence intervals cross zero. W7-X multigrid has a paired 95% interval of 21.77–23.45% lower wall time.

Convergence is a residual and fixed-point claim

v1.2 local result	Stage counts	FSQR	FSQZ	FSQL	Replay passes
Solovev	235 → 193 → 326	9.972×10^{-17}	5.409×10^{-18}	9.511×10^{-21}	1
W7-X	1315 → 1419 → 1372	9.997×10^{-13}	2.818×10^{-13}	1.299×10^{-13}	1

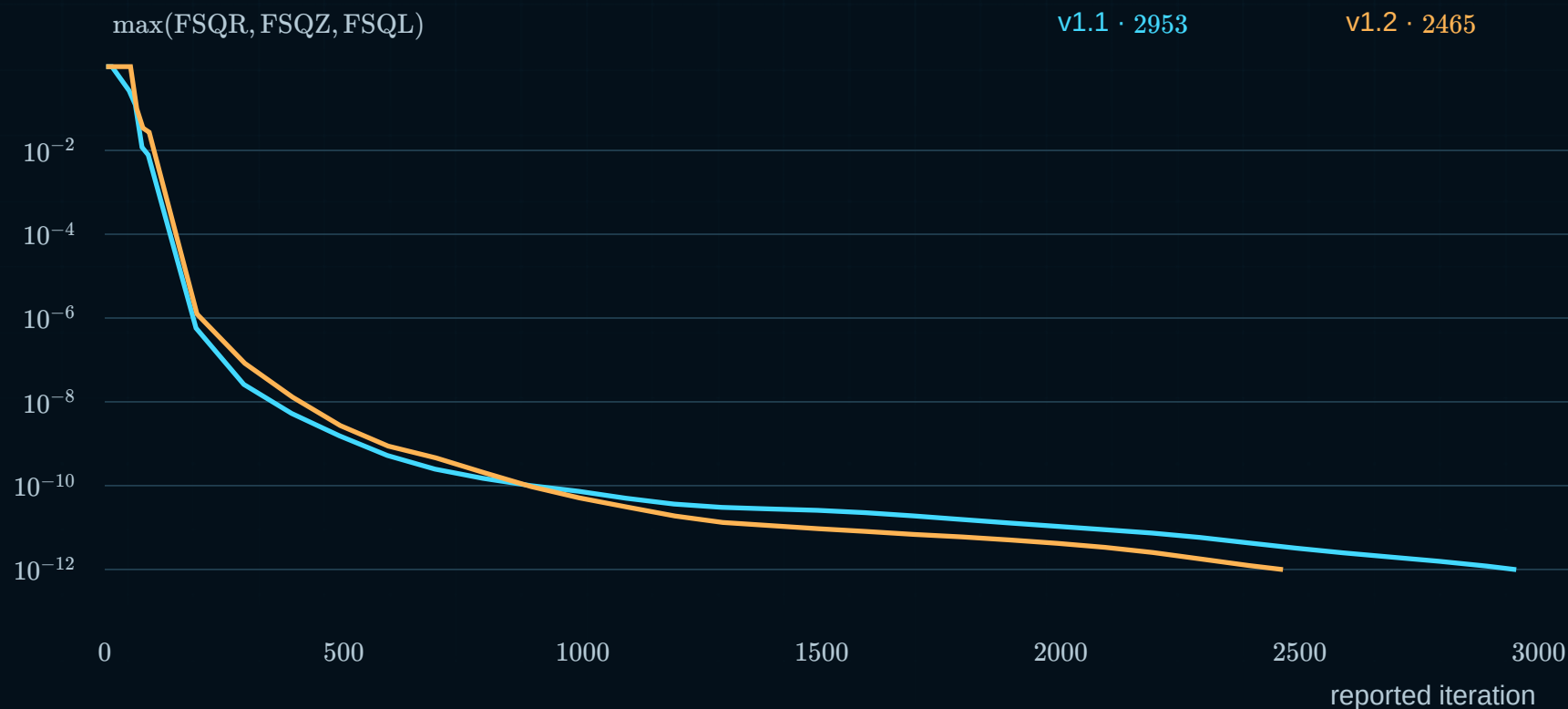
RE-EVALUATE THE RESULT

Save a final-grid checkpoint, then restart the same problem on the final grid. Both reproduced cases converge after one evaluation.

USE THE RIGHT EQUIVALENCE TEST

Seeds and transfer intentionally change the path. Different λ -gauge representatives can meet the same force tolerances; byte identity to v1.1 is not required.

A shorter convergence path, sampled from the logs



W7-X · single $n_s = 99$ · same 10^{-12} tolerance. Lines join printed samples; early restart details between samples are not resolved.

Saved passes have different costs on different grids

W7-X stage	v1.1 passes	v1.2 passes	Passes removed	Grid size
Coarse	1,877	1,315	562	33 × 30 × 36
Middle	1,617	1,419	198	66 × 30 × 36
Fine	2,011	1,372	639	99 × 30 × 36

The fine grid contributes 639 of the 1,399 saved passes. A total-pass percentage alone cannot predict the exact wall-time reduction.

Numerical ideas rejected by the evidence

Candidate	Measured result	Why it was not retained
Repeated / periodic step recovery	Recurring invalid Jacobians; no convergence	One-shot recovery is the qualified policy
Damping floor 0.0025...0.08	No convergence within 5,000 iterations	Does not cure the early transient
Preconditioner cadence 10	Best 2,853 vs 2,953 reference	Other intervals regress; larger algorithm change
Fine free W7-X envelope 0.12	1,953 vs 1,831 reference	Resolution-limited 0.03 chosen
Full fixed-boundary predictor for free W7-X	$1,449 + 1,723 = 3,172$ passes	More work than 1,831 direct passes

Appendix: later work

A Later releases extend both themes: concurrent solves, coarse-start tuning, and precision-aware execution.

v1.3: very coarse 3-D seeds help QA and QH differently

Analytic center	Reference	Coarse seed −0.10	Fewer passes
QA	1178 → 252 → 118 = 1548	886 → 226 → 100 = 1212	21.7%
QH	445 → 302 → 361 = 1108	303 → 293 → 314 = 910	17.9%

Four-worker Jacobian · 3 runs	Before	After	Wall reduction
QA	0.827 s	0.658 s	20.4%
QH	0.693 s	0.687 s	0.8%

Sensitivity reuse solves a different performance problem

RETAINED LINEARIZED EQUILIBRIUM

$$F_u \delta u = -F_x \delta x$$

Analytic CUDA JVPs and right-preconditioned restarted GMRES reuse the converged operator session for successive boundary directions.

REPORTED QUALIFICATION

QH residual-vector derivative difference: 5.2%.

Coordinate-invariant objective derivative difference: 0.14%.

No isolated timing is used here to claim faster nonlinear equilibrium convergence.

Current float branch: accuracy unlocks convergence

W7-X float · every stage $f_{\text{tol}} = 10^{-5}$	Result	Extra per-pass work
Absolute R_{00} quantization diagnostic	Double-evaluated FSQR 2.1083×10^{-4}	Precision floor evidence
Quantize displacement relative to edge	FSQR 4.8304×10^{-9} in the same diagnostic	Retain small radial structure
Reference storage alone	Final grid still stalls; best max residual 1.4070×10^{-5}	Does not complete cold convergence
Reference + poloidal float-float	$149 \rightarrow 277 \rightarrow 322$; FSQR 9.837748×10^{-6}	Fused into existing R/Z launches

These are 10^{-5} mixed-float accuracy experiments. They cannot be compared as speedups against the 10^{-12} double W7-X solve.

Cache the immutable float radius reference

TITAN Xp · 300 warmup + 500 measured	Before cache	After cache	Reduction
Reference / native float	541.90 μ s	525.55 μ s	3.02%
Reference / poloidal float-float	559.29 μ s	543.02 μ s	2.91%
Absolute-coefficient control	—	526.46 μ s	Same-session control

Compute the constant reference Fourier sum once per stage. One graph node disappears; no new buffer is allocated; the 149 → 277 → 322 trajectory is unchanged.

The gains that the evidence supports

EXECUTE EACH PASS EFFICIENTLY

RTX W7-X: 660.81 $\rightarrow$ 534.53 μ s.

RTX Solovev: 115.59 $\rightarrow$ 82.33 μ s.

Graphs matter most when submission is a large fraction of latency. Keep data resident and retain only measured kernel changes.

SPEND FEWER PASSES CONVERGING

W7-X multigrid: 5,505 $\rightarrow$ 4,106.

Solovev multigrid: 906 $\rightarrow$ 754.

Cold-start policy and smoother continuation reduce full force evaluations while meeting the configured residual tolerances.

The new TITAN Xp release comparison confirms the deterministic counts and measures complete wall time. Every headline points to commits, inputs, and raw or archived evidence.

Evidence and reproduction

Artifact	What it contains
evidence.md	Pinned source citations, measurement scopes, and limits
data/commit-audit.tsv	All 197 post-design-closeout commits and changed files
data/optimization-commits.md	42 highlighted optimization/support/evidence records
data/local/	Raw fixed-iteration JSON, CLI logs, exact inputs, GPU provenance
scripts/build_history.py → measure.py → summarize.py	Historical worktrees, controlled measurements, statistical summary
scripts/build_deck.py	Static HTML generation from narrative and measured results

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